Calculating CPTG Galaxy Rotation Curves

Resolved Polarization–Transport Rotation-Curve Solutions

UGC 00731, NGC 7331, and NGC 2955

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Abstract

This updated tutorial calculates three galaxy rotation curves using the newly resolved polarization–transport formulation of Curvature Polarization Transport Gravity (CPTG). UGC 00731, NGC 7331, and NGC 2955 span a gas-rich dwarf disk, an extended spiral, and a bulge-dominated spiral. Starting from the published SPARC gas, disk, and bulge contributions, the calculation constructs a signed baryonic acceleration, measures the local rotational support at each radius, and reconstructs a differentiable support profile. The latter supplies the curvature carrier, analytically differentiated polarization response, and nonnegative, unclipped transport coordinate. A nonlinear implicit equation is then solved for the radial gravitational acceleration and returned circular speed. Three measured radii in each galaxy are developed numerically, with the intermediate quantities exposed for independent calculation. The reconstructed field is conditioned by the observed velocities: $a_\star$ is calculated independently at every measured radius, while the three reduced coefficients are fitted once for each galaxy. This measured-field approach preserves the distinction between local gravitational organization, mathematical root closure, and observational velocity residuals. The post-solution Structural Mode N and its Curvature Structure Mode Index (CSMI) describe each solved field without entering the fit. A fixed-a_0 reference curve supplies a separate illustrative comparison. The method, radial examples, and complete-catalog summary use the same polarization–transport equation throughout.

1 Introduction

Galaxy rotation curves show how orbital speed varies with distance from a galaxy's center. The observed gas, stellar disk, and bulge distributions describe the material source of gravity, while measured circular speeds reveal the gravitational support required at each radius. Curvature Polarization Transport Gravity (CPTG) is a geometric extension of General Relativity: ordinary matter remains the physical source, but *curvature polarization* describes how the gravitational geometry reorganizes with field strength and structure, and *curvature transport* describes how that response is distributed through the system [4, 6]. Depending on that organization, a material distribution can exhibit stronger or weaker gravity. CPTG returns continuously to General Relativity when these additional responses become negligible; it does not require another gravitating matter component.

CPTG reconstructs each galaxy's measured gravitational field from the separately tabulated baryonic source and local rotational support. The structural acceleration $a_\star$ is determined independently at every observed radius rather than represented by one galaxy-wide constant; its precise role in the field equation is developed in Section 2 [6].

For orientation, Modified Newtonian Dynamics (MOND) supplies a familiar comparison using the fixed-a_0 radial-acceleration-relation (RAR) form in Eq. (10) [2, 3]. The resulting velocities, velocity-based χ^2, and separately evaluated RAR scatter are reported without entering the CPTG solution.

The galaxy equation is one realization of the same CPTG source–polarization–transport gravitational law used for galaxies, galaxy clusters, and black holes [6]. Galaxy support comes from rotation; cluster support comes from spherical pressure or Jeans reconstruction. Black holes require a relativistic covariant finite-state realization, not the disk equation applied inside a black hole. The physical coordinates change with the system, but the parent gravitational architecture remains the same.

The three worked examples cover UGC 00731 (12 radii), NGC 7331 (36 radii), and NGC 2955 (24 radii). Their distinct source distributions illustrate the same field equation; independent root checks and post-solution Structural Mode N complete the analysis.

2 The Resolved CPTG Galaxy Equation

The resolved law combines the signed baryonic source, measured structural acceleration, and differentiated support through the curvature carrier C_C, polarization response $B_{\rm pol}$, transport coordinate Σ_C^2, and nonlinear factor $\mathcal{S}$ [6].

At every observed radius, the measured local structural acceleration is

$$a_\star(r_i) = \frac{V_{\rm obs}^2(r_i)}{r_i}. \tag{1}$$

An uncertainty-weighted spline $g_{\rm sup,smooth}(r)$ supplies the radial derivatives needed by the

polarization operator. Its logarithmic slope determines the transport coordinate:

$$\Sigma_C^2(r) = \frac{1}{3}\left[1 - \frac{d\ln g_{\rm sup,smooth}}{d\ln r}\right], \qquad \Sigma_C^2(r) \geq 0. \tag{2}$$

The support reconstruction enforces $\Sigma_C^2 \geq 0$ without upper clipping; values above unity remain admissible. Smoothing supplies derivatives but does not replace measured $a_\star(r_i)$.

The resolved radial curvature invariant is

$$C_C(r) = g_{\rm sup,smooth}^2 + \kappa_C^2\left[\left(\frac{dg_{\rm sup,smooth}}{dr}\right)^2 + 2\left(\frac{g_{\rm sup,smooth}}{r}\right)^2\right]. \tag{3}$$

The registered radial polarization operator is

$$B_{\rm pol}(r) = \frac{1}{r^2}\left[(r^2+2\kappa_C^2)g_{\rm sup,smooth}C_C^{-2/3} - \frac{d}{dr}\left(r^2\kappa_C^2\frac{dg_{\rm sup,smooth}}{dr}C_C^{-2/3}\right)\right]. \tag{4}$$

The operator is the registered effective radial reduction used for disk galaxies, not a claim that a thin disk is a spherical mass distribution. The nonlinear response factor is

$$\mathcal{S}(g_C, r) = \left[1 + \left(\frac{g_C(r)}{a_\star(r)}\right)^{123/50}\right]^{-11/50}. \tag{5}$$

With direct baryonic acceleration g_b, the field equation is

$$g_C(r) = \frac{g_b(r) + A_C a_\star^{2/3}(r) B_{\rm pol}(r)\mathcal{S}(g_C, r)}{1 + c_C\Sigma_C^2(r)}. \tag{6}$$

Measured-field reconstruction. The radial gravitational field $g_C(r_i)$ is reconstructed from the signed baryonic source, local $a_\star(r_i)$, and differentiated support profile. The reduced coefficients κ_C, A_C, and c_C are fitted once per galaxy, not independently at each radius; the same implicit equation is then solved throughout the measured profile. Its equation-root residual tests mathematical closure, while velocity residuals quantify observational agreement. The coefficients have SI dimensions $[\kappa_C] = L$, $[A_C] = L^{2/3}T^{-4/3}$, and $[c_C] = 1$. Structural Mode N is read only after the solution.

3 Input Data and Unit Conversion

The inputs are the SPARC mass-model columns r, $V_{\rm obs}$, σ_V, $V_{\rm gas}$, $V_{\rm disk}$, and $V_{\rm bulge}$ [1]. Gas entries can be negative, representing signed contributions in the tabulation. That sign is preserved in the source. Numerical values are converted before solving:

$$r_{\rm m} = r_{\rm kpc}(3.085677581491367 \times 10^{19}\,{\rm m}), \qquad V_{\rm m/s} = 10^3 V_{\rm km/s}. \tag{7}$$

The three input profiles are treated independently, with their existing radii and tabulated velocity uncertainties unchanged. The adopted SPARC input distances are 12.5 Mpc (UGC 00731), 14.7 Mpc (NGC 7331), and 97.9 Mpc (NGC 2955). No observed baryonic source component is inferred from the solved gravitational pull.

4 Constructing the Direct Baryonic Source

The resolved calculation uses unweighted SPARC gas, disk, and bulge contributions. The signed-square convention is

$$Q(V) = V|V|. \tag{8}$$

The signed squared-velocity sum and direct acceleration are

$$V_b^{2,\mathrm{signed}}(r) = Q(V_{\mathrm{gas}}) + Q(V_{\mathrm{disk}}) + Q(V_{\mathrm{bulge}}), \qquad g_b(r) = \frac{V_b^{2,\mathrm{signed}}(r)}{r}. \tag{9}$$

The sign belongs to the radial gravitational contribution, not to a negative mass. A negative gas component acts outward; a negative *net* contribution also remains negative rather than being set to zero. The signed source is not the complete CPTG response. No extra gas, disk, or bulge weighting and no fitted stellar mass-to-light scaling is introduced. For comparison only, the fixed-a_0, unit-source MOND/RAR-form curve uses the same tabulated baryonic components with their unit-source normalization and the separate acceleration response

$$g_{\mathrm{RAR}} = \frac{g_b}{1 - \exp[-\sqrt{g_b/a_0}]}, \qquad a_0 = 1.2 \times 10^{-10}\ \mathrm{m\,s^{-2}}. \tag{10}$$

For this comparison, g_b is restricted to the nonnegative domain required by the square-root formula; this restriction does not alter the signed CPTG input. The expression uses the radial-acceleration-relation functional form of McGaugh, Lelli, and Schombert [3], with a_0 fixed rather than estimated separately for each galaxy. Its circular speed is $V_{\mathrm{MOND}} = \sqrt{r g_{\mathrm{RAR}}}$. This explicitly defined MOND/RAR-form comparator does not participate in the CPTG fit, support-state recovery, or Structural Mode calculation; it simply provides a common familiar reference for the green curves, worked-radius values, and reported residuals.

5 Solving the Implicit Polarization–Transport Equation

5.1 Step 1: Construct the Local Support State

For each tabulated radius first calculate $a_\star = V_{\mathrm{obs}}^2/r$ and its propagated velocity contribution $\sigma_g = 2|V_{\mathrm{obs}}|\sigma_V/r$. With $x = \ln r$ and $y = \ln a_\star$, fit an uncertainty-weighted cubic smoothing spline to $y(x)$, with weights $a_\star/\sigma_g$ and smoothing target equal to the number of sampled radii. The spline gives a continuous $g_s = g_{\mathrm{sup,smooth}}$ and its first two logarithmic derivatives. If the fitted angular velocity is not nonincreasing, constrain the support spline to make Σ_C^2 nonnegative, without placing an upper limit on it. The observed $a_\star$ values remain unchanged.

For evaluation of the radial operator, define

$$s = \frac{d\ln g_s}{d\ln r}, \qquad t = \frac{d^2\ln g_s}{d(\ln r)^2}, \qquad \frac{dg_s}{dr} = \frac{g_s s}{r}, \qquad \frac{d^2 g_s}{dr^2} = \frac{g_s}{r^2}(s^2 - s + t). \tag{11}$$

The three-radius tables expose s, t, and the resulting $\Sigma_C^2 = (1-s)/3$; the accompanying 72-radius dataset records the complete evaluated profiles.

5.2 Step 2: Evaluate Polarization and Transport

Use the fitted κ_C and the local g_s, s, t to calculate C_C and $B_{\rm pol}$. Rather than interpolating a second polarization-flux spline, the derivative in Eq. (4) is evaluated algebraically from the *same* support spline. Define

$$z = \left(\frac{\kappa_C}{r}\right)^2, \qquad h = s^2 + 2, \qquad C_C = g_s^2(1+hz). \tag{12}$$

The exact factored radial polarization derivative is

$$B_{\rm pol} = g_s C_C^{-2/3} \frac{Q_C(z)}{1+hz}, \tag{13}$$

where

$$\begin{aligned} Q_C(z) = 1 &+ \left(4 - s - t + \frac{4}{3}s^2\right) z \\ &+ \left[(s^2+2)(s-1)\left(\frac{s}{3} - 2\right) + t\left(\frac{s^2}{3} - 2\right)\right] z^2. \end{aligned} \tag{14}$$

Equations (12)–(14) are algebraically the same polarization operator as Eq. (4); their factored evaluation reduces cancellation at large κ_C/r. Having evaluated $B_{\rm pol}$, compute the denominator $D = 1 + c_C\Sigma_C^2$ and the source-independent response amplitude $P = A_C a_\star^{2/3} B_{\rm pol}$. Neither g_s nor the curvature operator is recalculated from the candidate solution g_C during the implicit root iteration.

5.3 Step 3: Solve and Verify the Implicit Field

The local root satisfies

$$F_i(g_{C,i}) = \left(1 + c_C\Sigma_{C,i}^2\right) g_{C,i} - g_{b,i} - A_C a_{\star,i}^{2/3} B_{{\rm pol},i} \mathcal{S}(g_{C,i}, r_i) = 0. \tag{15}$$

The galaxy-level search uses logarithmic bounds $10^{-4} \le \kappa_C/({\rm kpc}) \le 10^7$, $10^{-30} \le A_C \le 10^5$ (SI reduction), and $10^{-10} \le c_C \le 10^3$. A 30×18 initial grid in κ_C and c_C is screened; up to 12 distinct seeds undergo bounded least-squares refinement with at most 300 residual evaluations each. These are numerical search bounds, not physical constants. The UGC 00731 solution reaches the κ_C lower bound; NGC 7331 reaches the c_C upper bound with κ_C close to its upper bound. Every trial requires a converged implicit root. The optimized observational statistic is

$$\chi^2 = \sum_i \left[\frac{V_{\rm CPTG}(r_i) - V_{\rm obs}(r_i)}{\sigma_V(r_i)}\right]^2, \qquad V_{\rm CPTG}(r_i) = \sqrt{g_{C,i} r_i}. \tag{16}$$

For the separately reported acceleration-space RAR scatter, define $g_{\rm obs,i} = V_{\rm obs,i}^2/r_i$ and $\sigma_{\rm RAR} = {\rm std}_i[\Delta_i]$ with $\Delta_i = \log_{10}(g_{\rm model,i}/g_{\rm obs,i})$, evaluated independently for CPTG and the fixed-a_0 MOND reference in dex. This statistic is distinct from velocity-based χ^2.

After obtaining g_C, check its normalized equation residual

$$\epsilon_{F,i} = \frac{|F_i|}{|(1 + c_C\Sigma_{C,i}^2)g_{C,i}| + |g_{b,i}| + |A_C a_{\star,i}^{2/3} B_{{\rm pol},i}\mathcal{S}_i|}. \tag{17}$$

Newton iteration uses at most 28 updates with relative-update tolerance 2×10^{-11} and normalized residual below 5×10^{-8}. If needed, up to 180 damped fixed-point iterations provide a fallback, followed by a direct consistency check. Independent bracketing verifies the positive roots. These tolerances govern numerical closure, not the gravitational coefficients.

5.4 Working a Positive Root by Hand

Once the support derivatives and galaxy coefficients are known, the implicit field can be solved by successive substitution. At the outermost measured radius of NGC 2955, $r = 35.43\,{\rm kpc}$, the measured and computed quantities are

$$\begin{aligned} a_\star &= 4.713353542 \times 10^{-11}, \quad g_b = 4.192725240 \times 10^{-11}, \\ P &= A_C a_\star^{2/3} B_{\rm pol} = 1.749370078 \times 10^{-10}, \quad D = 3.780270536, \end{aligned} \tag{18}$$

with accelerations in ${\rm m\,s^{-2}}$. Starting at $g^{(0)} = a_\star$, repeatedly calculate

$$g^{(n+1)} = \frac{g_b + P\left[1 + \left(g^{(n)}/a_\star\right)^{123/50}\right]^{-11/50}}{D}. \tag{19}$$

Iteration	$\boldsymbol{g^{(n)}}$ **(m s^{-2})**	$\boldsymbol{g^{(n+1)}}$ **(m s^{-2})**
0	4.713354×10^{-11}	5.082232×10^{-11}
1	5.082232×10^{-11}	4.998369×10^{-11}
2	4.998369×10^{-11}	5.017367×10^{-11}
3	5.017367×10^{-11}	5.013059×10^{-11}
4	5.013059×10^{-11}	5.014036×10^{-11}

Table 1: Direct fixed-point steps for the NGC 2955 outer-radius field. Further iteration approaches the same positive root obtained by the numerical solver.

The positive root is $g_C = 5.013855200 \times 10^{-11}\,{\rm m\,s^{-2}}$, giving $V_{\rm CPTG} = \sqrt{r g_C} = 234.124435\,{\rm km\,s^{-1}}$. Substituting this root into Eq. (15) closes the field equation. This fixed-point table demonstrates the arithmetic; the implemented solver primarily uses Newton updates and verifies the final residual independently.

6 Reading Structural Mode After the Solve

Structural Mode N is a dimensionless diagnostic of the organization of a solved gravitational field [7]. It is determined only after the gravitational field has been solved, is not a fit input, and is measured for each solved object. The registered galaxy sampled-data estimator uses $\kappa_{\mathrm{struct}} = 1\,\mathrm{kpc}$:

$$C_i^{\mathrm{str}} = g_{C,i}^2 + \kappa_{\mathrm{struct}}^2 (D_r g_C)_i^2, \qquad u_i = |(D_r C^{\mathrm{str}})_i|. \tag{20}$$

On radii $r_i > R/5$ with $R = \max_i r_i$, an equal-weight five-point moving average is applied to u_i if there are more than five retained samples; otherwise $w_i = u_i$. The resulting sampled readout is

$$\lambda_{\mathrm{num}} = \frac{\sum_i r_i w_i}{\sum_i w_i}, \qquad N_{\mathrm{num}} = \frac{R}{\lambda_{\mathrm{num}}}. \tag{21}$$

For sampled radii, D_r is calculated using centered nonuniform-spacing finite differences at interior points and one-sided differences at the endpoints. The five-point mean uses equal weights and zero extension outside the retained outer-radius sequence. The gradients of g_C and C^{str} are taken over the full radial array before selecting $r_i > R/5$. No Δr_i quadrature factor is inserted in this discrete estimator. Only after N_{num} is computed does the descriptive CSMI category follow from the registered numerical bands. The same readout is used for each independent solved field.

7 Example I: UGC 00731 as a Gas-Rich Dwarf Disk

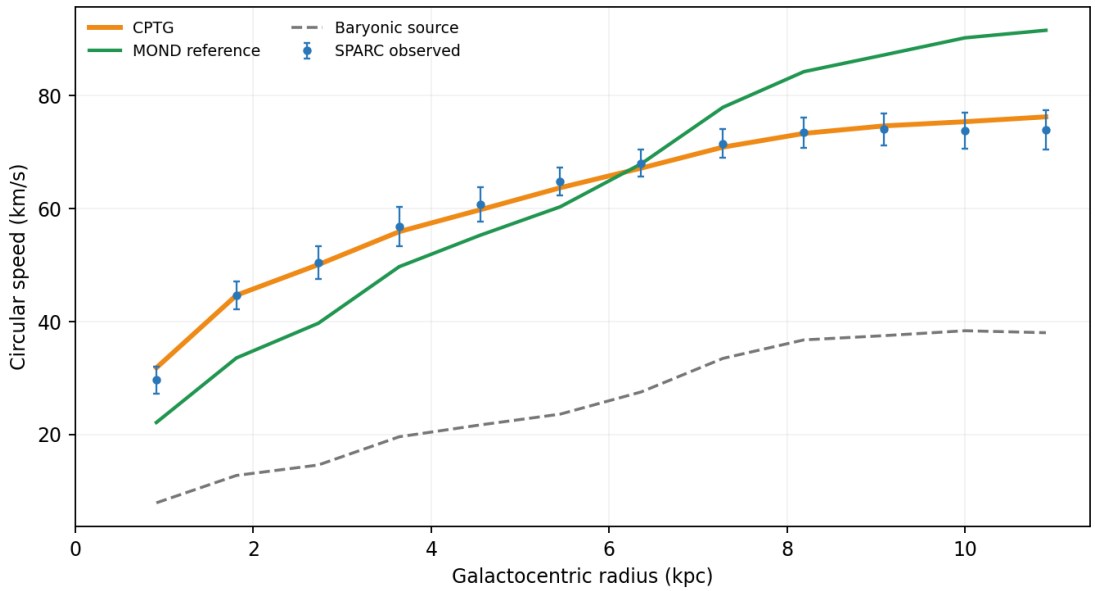


Figure 1: Resolved rotation-curve calculation for UGC 00731. Blue markers and error bars are the observed SPARC velocities; the orange curve is the resolved CPTG solution, the green curve the fixed-a_0 comparison defined in Eq. (10), and the dashed gray curve the direct material-source velocity.

The gas-rich outer profile provides a direct illustration of the response to a weak baryonic source.

7.1 Input Profile and Solved State

UGC 00731 is an extended, low-surface-brightness dwarf disk. In the adopted SPARC input its bulge entry vanishes at every radius, and the gas contribution increasingly dominates the direct source toward the outside. This exposes the polarization–transport response in a low-source environment.

The adopted SPARC profile contains 12 data points, from 0.91 to 10.91 kpc. The three fitted resolved-field quantities are

$$\kappa_C = 1.00000\times10^{-4}\ \text{kpc}, \qquad A_C = 6.62318\times10^{-7}, \qquad c_C = 12.610943. \tag{22}$$

7.2 Following the Calculation Across Three Radii

The three-radius table follows the calculation from observed inputs through baryonic source, support derivatives, polarization, transport, and returned circular speeds. Full-profile χ^2 and RAR scatter appear beneath the worked radii.

Quantity	0.91 kpc	6.36 kpc	10.91 kpc
Observed and baryonic-source velocities (km s^{-1})			
$V_{\rm obs}$	29.60	68.00	73.90
σ_V	2.38	2.38	3.42
$V_{\rm gas}$	0.80	23.38	35.75
$V_{\rm disk}$	7.82	14.47	12.83
$V_{\rm bulge}$	0.00	0.00	0.00
Local support and direct baryonic acceleration (m s^{-2})			
$a_\star$	3.120×10^{-11}	2.356×10^{-11}	1.622×10^{-11}
g_b	2.201×10^{-12}	3.852×10^{-12}	4.285×10^{-12}
$g_{\rm sup,smooth}$	3.247×10^{-11}	2.365×10^{-11}	1.692×10^{-11}
Differentiated support, polarization, and transport			
s	0.102289	-0.500643	-0.744841
t	-0.198621	-0.421571	-0.483451
Σ_C^2	0.299237	0.500214	0.581614
$B_{\rm pol}$	3.135×10^3	3.484×10^3	3.895×10^3
$\mathcal{S}$	0.823496	0.864497	0.843861
$D=1+c_C\Sigma_C^2$	4.773659	7.308173	8.334696
$P\mathcal{S}$ (m s^{-2})	1.694×10^{-10}	1.639×10^{-10}	1.395×10^{-10}
g_C (m s^{-2})	3.596×10^{-11}	2.296×10^{-11}	1.725×10^{-11}
Returned circular speeds (km s^{-1})			
$V_{\rm CPTG}$	31.78	67.13	76.21
$V_{\rm MOND}$	22.09	67.89	91.53
Full-profile statistics (12 radii)			
$\chi^2_{\rm CPTG}$	2.142874		
$\chi^2_{\rm MOND}$	153.177266		
$\sigma_{\rm RAR,CPTG}$ (dex)	0.021482		
$\sigma_{\rm RAR,MOND}$ (dex)	0.156053		

Table 2: UGC 00731 at three observed radii, evaluated with the same source, differentiated support, polarization, transport, and implicit field equations. The listed s and t are the support-spline logarithmic derivatives used to obtain $B_{\rm pol}$.

7.3 Outer-Radius Calculation and Structural Readout

At $r = 10.91$ kpc the gas-rich source is only $g_b = 4.285408\times10^{-12}\,\mathrm{m\,s^{-2}}$. The differentiated support gives $\Sigma_C^2 = 0.581614$ and transport denominator $D = 8.334696$. Combining the tabulated suppressed polarization contribution with the baryonic source gives

$$g_C = \frac{g_b + P\mathcal{S}}{D} = 1.725162\times10^{-11}\ \mathrm{m\,s^{-2}}, \tag{23}$$

corresponding to $V_{\rm CPTG} = 76.21\,\mathrm{km\,s^{-1}}$ at this radius. The full-profile velocity residuals have RMS 1.190274 and mean absolute value 0.966265 km s^{-1}; their median absolute fractional value is 1.422%, with 12/12 radii within 10%. After solving, $N_{\rm num} = 1.880747932$, $\lambda_{\rm num} = 5.800884$ kpc, and CSMI is LSB Dwarf Disk. The complete χ^2 and RAR values are given in Table 2.

8 Example II: NGC 7331 as an Extended Spiral Disk

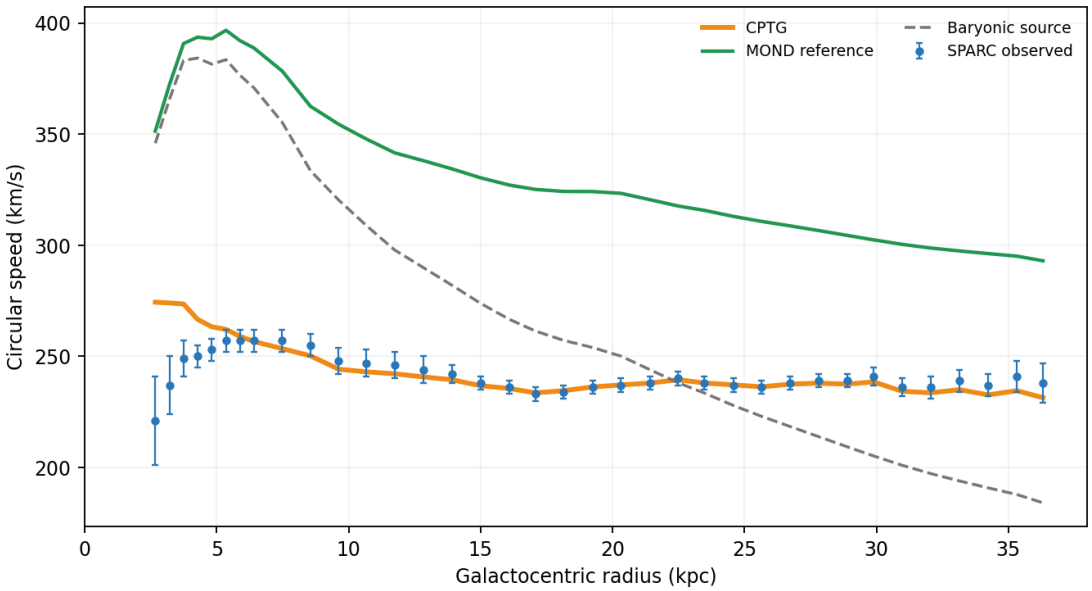


Figure 2: Resolved rotation-curve calculation for NGC 7331. Blue markers and error bars are the observed SPARC velocities; the orange curve is the resolved CPTG solution, the green curve the fixed-a_0 comparison defined in Eq. (10), and the dashed gray curve the direct material-source velocity.

The extended disk and bulge profile tests the same radial operations across a wide range of source contributions.

8.1 Input Profile and Solved State

NGC 7331 is an extended spiral with a strong disk contribution and a separately tabulated stellar bulge. The measured circular speed is comparatively flat over much of the resolved outer region. Its extended radial coverage makes it a useful example of how the disk, bulge, and gas sources enter the same polarization–transport calculation.

The adopted SPARC profile contains 36 data points, from 2.67 to 36.31 kpc. The three fitted resolved-field quantities are

$$\kappa_C = 9.62178\times10^{6}\ \text{kpc}, \qquad A_C = 1.21588\times10^{-8}, \qquad c_C = 1000.000000. \tag{24}$$

8.2 Following the Calculation Across Three Radii

Table 3 applies the same sequence to the extended spiral, including the signed inner gas contribution.

Quantity	2.67 kpc	18.15 kpc	36.31 kpc
Observed and baryonic-source velocities (km s^{-1})			
$V_{\rm obs}$	221.00	234.00	238.00
σ_V	20.00	3.00	9.00
$V_{\rm gas}$	-12.02	53.93	47.97
$V_{\rm disk}$	317.90	245.89	173.78
$V_{\rm bulge}$	136.84	52.49	37.11
Local support and direct baryonic acceleration (m s^{-2})			
$a_\star$	5.928×10^{-10}	9.777×10^{-11}	5.056×10^{-11}
g_b	1.452×10^{-9}	1.181×10^{-10}	3.024×10^{-11}
$g_{\rm sup,smooth}$	6.351×10^{-10}	1.004×10^{-10}	5.320×10^{-11}
Differentiated support, polarization, and transport			
s	-0.351959	-1.099568	-0.670735
t	-1.130637	0.350493	0.886366
Σ_C^2	0.450653	0.699856	0.556912
$B_{\rm pol}$	6.463×10^{7}	3.105×10^{7}	1.838×10^{7}
$\mathcal{S}$	0.741369	0.857512	0.871313
$D=1+c_C\Sigma_C^2$	451.652920	700.855888	557.911550
$P\mathcal{S}$ (m s^{-2})	4.111×10^{-7}	6.872×10^{-8}	2.663×10^{-8}
g_C (m s^{-2})	9.135×10^{-10}	9.821×10^{-11}	4.778×10^{-11}
Returned circular speeds (km s^{-1})			
$V_{\rm CPTG}$	274.34	234.53	231.38
$V_{\rm MOND}$	351.35	324.20	292.98
Full-profile statistics (36 radii)			
$\chi^2_{\rm CPTG}$	48.901251		
$\chi^2_{\rm MOND}$	18286.363158		
$\sigma_{\rm RAR,CPTG}$ (dex)	0.041872		
$\sigma_{\rm RAR,MOND}$ (dex)	0.067503		

Table 3: NGC 7331 at three observed radii, evaluated with the same source, differentiated support, polarization, transport, and implicit field equations. The listed s and t are the support-spline logarithmic derivatives used to obtain $B_{\rm pol}$.

8.3 Extended-Disk Response and Structural Readout

The inner tabulated gas contribution at 2.67 kpc is negative and remains signed in the source. At 36.31 kpc, the differentiated support gives $D = 557.911550$; the implicit solution $g_C = 4.778298 \times 10^{-11}\,\mathrm{m\,s^{-2}}$ yields $V_{\rm CPTG} = 231.38\,\mathrm{km\,s^{-1}}$. The large fitted κ_C and the transport coefficient at its search bound are recorded above rather than assigned at individual radii.

Across the full 36-radius profile, the RMS and mean absolute velocity residuals are 12.328793 and 5.848357 km s^{-1}, the median absolute fractional residual is 1.036%, and 34/36 radii differ by less than 10%. The post-solution structural readout is $N_{\rm num} = 3.068560742$, $\lambda_{\rm num} = 11.832909$ kpc, CSMI Early Spiral. Table 3 reports both velocity χ^2 and separate RAR scatter.

9 Example III: NGC 2955 as a Centrally Concentrated Spiral

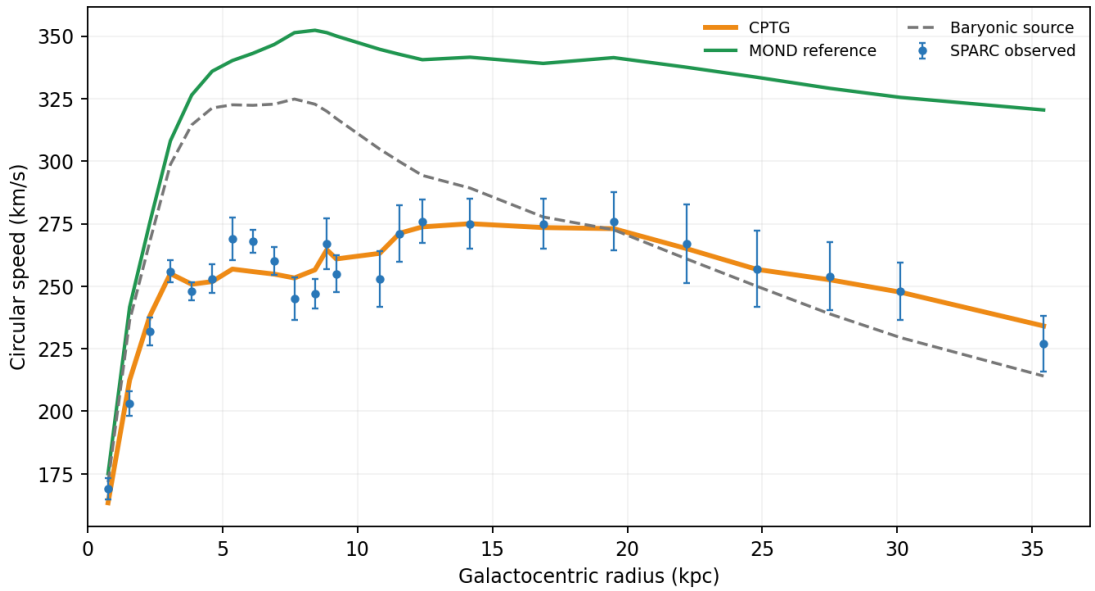


Figure 3: Resolved rotation-curve calculation for NGC 2955. Blue markers and error bars are the observed SPARC velocities; the orange curve is the resolved CPTG solution, the green curve the fixed-a_0 comparison defined in Eq. (10), and the dashed gray curve the direct material-source velocity.

The centrally concentrated source and outer transport behavior make this case especially useful for inspecting the completed implicit field.

9.1 Input Profile and Solved State

NGC 2955 provides the contrasting centrally concentrated case. The bulge component dominates the inner direct source, while the disk and gas contributions change with radius. The same local support, polarization, transport, and implicit closure sequence is evaluated at every measured radius.

The adopted SPARC profile contains 24 data points, from 0.75 to 35.43 kpc. The three fitted resolved-field quantities are

$$\kappa_C = 2.23259\times10^{0}\ \text{kpc}, \qquad A_C = 4.73760\times10^{-7}, \qquad c_C = 2.460628. \tag{25}$$

9.2 Following the Calculation Across Three Radii

Table 4 records the concentrated inner source and the outer transport coordinate; the latter is examined below.

Quantity	0.75 kpc	8.86 kpc	35.43 kpc
Observed and baryonic-source velocities (km s^{-1})			
$V_{\rm obs}$	169.00	267.00	227.00
σ_V	4.27	10.30	11.20
$V_{\rm gas}$	0.00	32.44	77.83
$V_{\rm disk}$	53.73	258.41	176.46
$V_{\rm bulge}$	163.04	185.85	92.96
Local support and direct baryonic acceleration (m s^{-2})			
$a_\star$	1.234×10^{-9}	2.608×10^{-10}	4.713×10^{-11}
g_b	1.273×10^{-9}	3.744×10^{-10}	4.193×10^{-11}
$g_{\rm sup,smooth}$	1.231×10^{-9}	2.453×10^{-10}	4.631×10^{-11}
Differentiated support, polarization, and transport			
s	-0.735441	-0.896279	-2.389708
t	1.089478	0.678763	-2.987227
Σ_C^2	0.578480	0.632093	1.129903
$B_{\rm pol}$	3.197×10^{3}	1.678×10^{3}	2.830×10^{3}
$\mathcal{S}$	0.873697	0.862726	0.843790
$D=1+c_C\Sigma_C^2$	2.423425	2.555346	3.780271
$P\mathcal{S}$ (m s^{-2})	1.523×10^{-9}	2.800×10^{-10}	1.476×10^{-10}
g_C (m s^{-2})	1.154×10^{-9}	2.561×10^{-10}	5.014×10^{-11}
Returned circular speeds (km s^{-1})			
$V_{\rm CPTG}$	163.40	264.60	234.12
$V_{\rm MOND}$	175.07	351.39	320.48
Full-profile statistics (24 radii)			
$\chi^2_{\rm CPTG}$	22.773087		
$\chi^2_{\rm MOND}$	2737.707507		
$\sigma_{\rm RAR,CPTG}$ (dex)	0.021454		
$\sigma_{\rm RAR,MOND}$ (dex)	0.059656		

Table 4: NGC 2955 at three observed radii, evaluated with the same source, differentiated support, polarization, transport, and implicit field equations. The listed s and t are the support-spline logarithmic derivatives used to obtain $B_{\rm pol}$.

9.3 Unclipped Outer Transport and Structural Readout

At $r = 35.43$ kpc, $\Sigma_C^2 = 1.129903$ exceeds unity without clipping. The transport denominator is $D = 3.780271$, and the completed positive root $g_C = 5.013855\times10^{-11}\,\mathrm{m\,s^{-2}}$ gives $V_{\rm CPTG} = 234.12\,\mathrm{km\,s^{-1}}$. Section 5.4 shows the successive-substitution steps for this same radius; no second derivation is needed here.

The 24-radius RMS and mean absolute velocity residuals are 6.015581 and 4.583934 km s^{-1}, with median absolute fractional residual 1.105% and 24/24 radii within 10%. Post-solution measurements give $N_{\rm num} = 3.084615227$, $\lambda_{\rm num} = 11.486036$ kpc, and CSMI Early Spiral. The CSMI assignment follows the solved field rather than the input bulge morphology; Table 4 records the full-profile χ^2 and RAR scatter.

10 Cross-Case Summary

Table 5 places the three independently reconstructed fields side by side, including their 72 measured radii, velocity statistics, separately evaluated RAR scatter, root residuals, and post-solution structure.

Quantity	**UGC 00731**	**NGC 7331**	**NGC 2955**
Observed radial profile and fitted response			
Measured radii	12	36	24
Radial extent (kpc)	0.91–10.91	2.67–36.31	0.75–35.43
κ_C (kpc)	1.000×10^{-4}	9.622×10^{6}	2.233×10^{0}
A_C (SI)	6.623×10^{-7}	1.216×10^{-8}	4.738×10^{-7}
c_C	12.610943	1000.000000	2.460628
Observed-velocity fit and equation closure			
χ^2_{CPTG}	2.142874	48.901251	22.773087
RAR scatter, CPTG (dex)	0.021482	0.041872	0.021454
$\chi^2/(N_{\mathrm{data}}-3)$	0.238097	1.481856	1.084433
Maximum normalized field residual	7.83×10^{-17}	1.76×10^{-16}	1.20×10^{-16}
Fixed-a_0 reference			
χ^2_{MOND}	153.177266	18286.363158	2737.707507
RAR scatter, MOND (dex)	0.156053	0.067503	0.059656
Post-solution structural measurements			
N_{num}	1.880747932	3.068560742	3.084615227
λ_{num} (kpc)	5.800884	11.832909	11.486036
CSMI	LSB Dwarf Disk	Early Spiral	Early Spiral

Table 5: Velocity-based χ^2, separately calculated acceleration-space RAR scatter, polarization–transport coefficients, and post-solution structural readouts.

Across all 72 measured radii, $\chi^2_{\mathrm{CPTG}} = 73.817212$ and $\chi^2_{\mathrm{MOND}} = 21177.247931$. The pooled RAR residual scatters are 0.033415 dex (CPTG) and 0.137958 dex (MOND). Each galaxy is calculated independently; Structural Mode does not enter the fit.

11 Numerical Convergence and Physical Interpretation

Independent bracketing confirms a positive root at all 72 measured radii in the worked examples. Their largest archived normalized equation residual is below 2×10^{-16}. These checks establish numerical closure of the implemented implicit equations; the velocity-based statistics quantify agreement with the separate observed measurements.

The 153-galaxy primary-sample calculation verifies 3,168 positive measured-radius roots. Table 6 gives the catalog-wide velocity χ^2 and independently evaluated acceleration-space RAR statistics. Solver tolerances and fallback procedures are specified in Section 5.

Primary-sample statistic	**CPTG**	**MOND reference**
Galaxies / measured radii	153 / 3,168	153 / 3,168
Total velocity χ^2	16,472.292025	687,218.159242
Pooled RAR scatter (dex)	0.102497	0.275381
Pooled RAR mean log residual (dex)	0.022238	0.171442
Pooled RAR RMS log residual (dex)	0.104882	0.324388
Mean galaxy RAR scatter (dex)	0.069075	0.115958

Table 6: Full 153-galaxy SPARC primary sample. Pooled RAR statistics use all 3,168 valid logarithmic acceleration residuals; mean galaxy scatter averages each galaxy's separately evaluated scatter. The fixed-a_0 MOND values are included only as a reference.

12 Relation to the Earlier Galaxy-Limit Formulation

The earlier tutorial, *Calculating CPTG Galaxy Rotation Curves: DDO-154, NGC-3198, and NGC-7814* [5], describes a single-structural-scale reduction with fixed source coefficients and a derived transport radius. That formulation remains a documented galaxy-limit realization of the parent law. Here the directly tabulated gas, disk, and bulge source contributions are used without extra weights, $a_\star$ is evaluated independently at each measured radius, and differentiated support determines the resolved polarization and transport response. The three reduced coefficients belong to the resolved galaxy fit. Structural Mode N is a post-solution measurement in both formulations.

Table 7 reports the two full-primary-sample records for the same 153 galaxies and 3,168 radial measurements [6]. Because source mapping, support reconstruction, and polarization–transport reduction differ, the totals characterize each complete method rather than any single coefficient change.

CPTG galaxy formulation	**Galaxies**	**Radii**	**χ^2 total**
Earlier single-scale galaxy limit	153	3,168	27,377.456854
Resolved support-conditioned field	153	3,168	16,472.292025
Difference in χ^2	—	—	**10,905.164829**

Table 7: Primary-sample SPARC results for the earlier galaxy reduction and the resolved polarization–transport formulation. Values are reproduced to a consistent six-decimal reporting precision from the full-sample numerical records.

13 Galaxy Analysis with the CPTG SPARC Browser Workbench

The CPTG SPARC Browser Workbench v1.13.1 is an interactive scientific application for calculating and examining galaxy rotation curves. Using published SPARC measurements and baryonic mass models, it analyzes individual galaxies or batches from the primary sample through the same resolved CPTG procedure developed in this tutorial.

For each galaxy, the interface displays the signed baryonic source, measured local structural acceleration, differentiated support, polarization, and transport alongside the fitted coefficients and solved velocities. Post-solution Structural Mode N, structural scale λ_{num}, and CSMI classification are reported separately, allowing the gravitational reconstruction to be examined at each radius.

The browser provides searchable galaxy selection, individual and batch execution, interactive rotation-curve plots, and detailed numerical results. Observed SPARC velocities appear alongside CPTG solutions and the familiar fixed-a_0 MOND reference. Separate velocity statistics include residuals, χ^2, and χ^2 per measured point for both calculations.

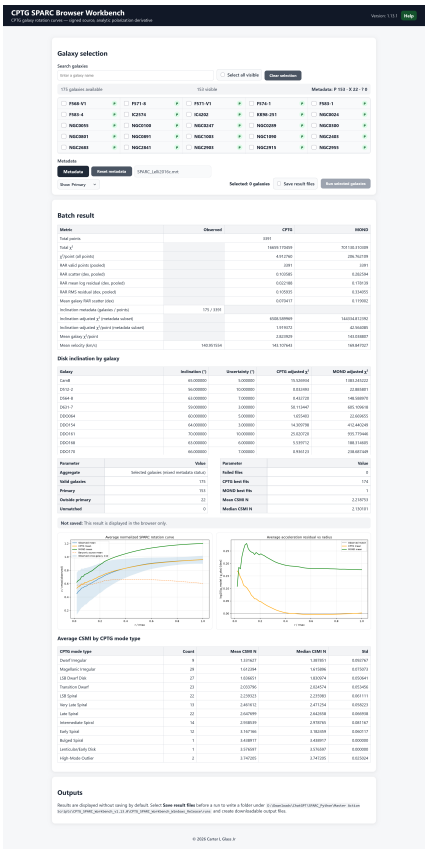


Figure 4: The CPTG SPARC Browser Workbench, showing batch-analysis results, sample statistics, and structural measurements.

The Radial Acceleration Relation (RAR) diagnostics are evaluated separately from velocity-based χ^2. At each valid radius, the observed acceleration $g_{\text{obs}} = V_{\text{obs}}^2/r$ is compared with each calculated acceleration. The logarithmic residual $\log_{10}(g_{\text{model}}/g_{\text{obs}})$ supplies the RAR scatter in dex, mean residual, RMS residual, and valid-point count for CPTG and MOND independently. These acceleration-space measurements do not alter either gravitational calculation.

Official SPARC metadata provide galaxy classification, distance, disk inclination, inclination uncertainty, and sample membership. Where available, inclination-adjusted velocity statistics are displayed separately without modifying the standard CPTG solution.

Batch summaries retain individual galaxy results while reporting successful and failed counts, primary-sample coverage, aggregate velocity χ^2, pooled RAR statistics, mean galaxy RAR scatter, and Structural Mode measurements. Full-precision radius-level exports, rotation-curve figures, and downloadable result packages support reproduction of the worked examples and further SPARC analysis [8].

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